

Research Idea: BHR_71, HH_212, IRAS_0416+2706,
L1551_IRS_5, CG_30, HH_46, GAL_331.5-00.1,
IRAS_04166+2706

ALMA Thesis Planner (automated pipeline)

0.1 A uniform two-transition SiO excitation study of the inner outflow regions in the SMORES ACA survey (2017.1.01053.S)

One-sentence question. Across the seven SMORES outflows, does SiO emission arise from a kinematically distinct, more highly excited gas component than the bulk CO-traced entrained outflow — and does the SiO excitation temperature and column density vary systematically with a self-measured outflow property (maximum CO velocity extent) rather than with externally assumed source ages?

Why this is the tractable core of the dataset. The proposal’s headline claim is spatio-chemical differentiation between jet, shocked wall, and ambient gas. A full test of that claim across 5 tunings $\times$ 42 member-OUS rows $\times$ dozens of species is a multi-year paper. But the survey happens to contain *two* transitions of the same shock tracer — the annotation lists **SiO J=7–6 at 303.927 GHz** (falling in the 302.0–304.0 GHz window) and **SiO J=8–7 at 347.331 GHz** (falling in the 345.2–349.1 GHz windows), together with **$^{12}\text{CO J=3–2}$ at 345.796 GHz** in the same tuning as SiO 8–7. Both tunings exist for all seven targets, so a two-point excitation analysis can be done on a strictly common footing with only ~ 14 of the 42 archive rows. Note that this line list is explicitly flagged in the data description as *generated from general knowledge, to be verified against Splatalogue before imaging*; the description also states that because these are Galactic sources within a few hundred pc, no redshift correction is needed and rest frequencies are used directly. Confirming those rest frequencies (and the exact SiO 7–6 / 8–7 / CO 3–2 quantum-number assignments and Einstein A / E_upper values) in Splatalogue or CDMS is step one of the thesis, not an assumption.

Scope-defining feasibility constraint. These are single-pointing ACA observations (‘Mosaic: no’), so the usable field is roughly the 7 m primary beam — of order half an arcminute at Band 7. With 3.0–4.6'' synthesized beams, that is only a handful of independent resolution elements per source. The project is therefore deliberately *aperture-based spectroscopy of the inner outflow*, not extended outflow mapping: three apertures per source (continuum/driving-source position, blue-shifted lobe, red-shifted lobe, all inside the primary-beam FWHM), which is exactly the number of measurements a single student can extract, error-bar, and defend.

Work plan (semester-scale). 1. *Data retrieval and audit.* Download QA2 products for the two relevant tunings for all seven targets from the ALMA archive (‘2017.1.01053.S’). Audit the target list first: the archive rows list both ‘IRAS_0416+2706’ and ‘IRAS_04166+2706’ at identical coordinates, and three targets sit at northern declinations despite the abstract’s “seven southern outflows” wording — the student should reconcile names and pointing centres before merging

anything. 2. *Homogenization.* Convolve all cubes to a single common circular beam ($\approx 5''$, set by the coarsest listed resolution of $4.582''$), regrid to a common spatial grid and to the coarser common velocity channel (the two tunings are listed at 1.078 and 0.943 km/s), and verify flux-scale consistency. This step, done properly, is the thesis’s real methodological contribution: any ratio computed from natively differing beams is meaningless. 3. *Measurement.* Extract spectra in the three apertures per source; fit or integrate SiO 7–6, SiO 8–7, and CO 3–2 over matched velocity intervals, defining low- and high-velocity intervals from the CO 3–2 line profile itself. 4. *Analysis.* From the SiO 8–7 / 7–6 integrated-intensity ratio, derive an LTE excitation temperature and SiO column density per aperture (optically thin, single- T_{ex} assumption, stated explicitly with its failure modes). Compare high-velocity to low-velocity intervals within each source, and rank sources by CO 3–2 maximum velocity extent — a quantity measured from these cubes alone — to test for a trend. 5. *Robustness.* Propagate the quoted line sensitivities (7.2–20.2 mJy/beam at 10 km/s) into real upper limits. Non-detections are a legitimate result here and are reported as limits, not dropped.

Deliverable. A homogenized, publicly reproducible set of matched-beam SiO and CO moment maps and aperture spectra for seven outflows, plus a table of SiO T_{ex} , $N(\text{SiO})$, and $N(\text{SiO})/N(\text{CO})$ proxies with uncertainties — enough to state whether SiO in these sources is confined to a distinct high-velocity, warmer component, and whether that component’s properties track outflow velocity extent.

Honest caveats built into the scope. Physical-scale and mass conversions require adopted literature distances, which the student must source and cite explicitly; the description gives only “within a few hundred pc,” so all primary results are framed as distance-independent line ratios, with any AU-scale or mass statement clearly flagged as distance-dependent. CO 3–2 is almost certainly optically thick and is used as a kinematic template and qualitative entrained-gas reference, not for absolute column densities. A two-transition ratio constrains excitation only weakly; the thesis should say so and treat the result as a first-order T_{ex} estimate. If SiO proves undetected in the fainter targets at ACA sensitivity, the fallback — already scientifically worthwhile — is a detections-plus-limits census with the same homogenization machinery, optionally extended to the SO 344.311 GHz / CH₃OH 338.409 GHz lines listed in the annotation (again, frequencies to be Splatalogue-verified) as a secondary shock-chemistry check.